



Cambridge International AS & A Level

CANDIDATE
NAME

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MATHEMATICS

9709/31

Paper 3 Pure Mathematics 3

May/June 2023

1 hour 50 minutes

You must answer on the question paper.

You will need: List of formulae (MF19)

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- If additional space is needed, you should use the lined page at the end of this booklet; the question number or numbers must be clearly shown.
- You should use a calculator where appropriate.
- You must show all necessary working clearly; no marks will be given for unsupported answers from a calculator.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.

INFORMATION

- The total mark for this paper is 75.
- The number of marks for each question or part question is shown in brackets [].

This document has **20** pages. Any blank pages are indicated.



$$3e^{2x} - 4e^{-2x} = 5.$$

[3]

[illegible]

- 2 (a) Sketch the graph of $y = |2x + 3|$.

[1]

- (b) Solve the inequality $3x + 8 > |2x + 3|$.

[3]

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- 4 (a) Show that the equation $\sin 2\theta + \cos 2\theta = 2 \sin^2 \theta$ can be expressed in the form

$$\cos^2 \theta + 2 \sin \theta \cos \theta - 3 \sin^2 \theta = 0. \quad [2]$$

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- (b) Hence solve the equation $\sin 2\theta + \cos 2\theta = 2 \sin^2 \theta$ for $0^\circ < \theta < 180^\circ$. [4]

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(a) Show that $\frac{dy}{dx} = \frac{2xy}{2ay - x^2}$. [4]

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[4]

[illegible]

- $$\overrightarrow{OA} = \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix}, \quad \overrightarrow{OB} = \begin{pmatrix} 4 \\ 3 \\ 2 \end{pmatrix} \quad \text{and} \quad \overrightarrow{OC} = \begin{pmatrix} 3 \\ -2 \\ -4 \end{pmatrix}.$$

(a) Find the position vector of D .

[3]

This image shows a full page of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page, typical of notebook or legal stationery. There are no margins, text, or other markings on the page.

- (b) The angle between BA and BC is θ .

Find the exact value of $\cos \theta$.

[3]

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- (c) Hence find the area of $ABCD$, giving your answer in the form $p\sqrt{q}$, where p and q are integers.

[4]

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- 7 The variables x and y satisfy the differential equation

$$\cos 2x \frac{dy}{dx} = \frac{4 \tan 2x}{\sin^2 3y},$$

where $0 \leq x < \frac{1}{4}\pi$. It is given that $y = 0$ when $x = \frac{1}{6}\pi$.

Solve the differential equation to obtain the value of x when $y = \frac{1}{6}\pi$. Give your answer correct to 3 decimal places. [8]

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[illegible]

[5]

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- (b)** Hence find the exact value of $\int_0^4 f(x) \, dx$, giving your answer in the form $a + b \ln c$, where a , b and c are integers. [5]

[illegible]

(a) Show that $a = \frac{1}{2} \ln(4a + 2)$. [5]

[illegible]

- (b) Verify by calculation that a lies between 0.5 and 1.

[2]

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- (c) Use an iterative formula based on the equation in (a) to determine a correct to 2 decimal places. Give the result of each iteration to 4 decimal places.

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10 The polynomial $x^3 + 5x^2 + 31x + 75$ is denoted by $p(x)$.

(a) Show that $(x + 3)$ is a factor of $p(x)$. [2]

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(b) Show that $z = -1 + 2\sqrt{6}i$ is a root of $p(z) = 0$. [3]

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(c) Hence find the complex numbers z which are roots of $p(z^2) = 0$.

[7]

[illegible]

Additional Page

If you use the following lined page to complete the answer(s) to any question(s), the question number(s) must be clearly shown.

[illegible]

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